

PROJECT PROPOSAL

AI-Driven Civic Pothole Detection & Spatial Management System

Prepared By:

Muhammad Rehan Siddiqui

Abdullah Razzaq

Ariz Ali

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1 Project Overview

The AI-Driven Civic Pothole Detection and Spatial Management System is a comprehensive web-based platform designed to automate the triage, reporting, and tracking of road infrastructure damage. By crowdsourcing user-submitted data and placing an Artificial Intelligence (AI) verification model at the gateway of data entry, the system filters out invalid reports. Furthermore, it leverages advanced spatial database mapping to organize reports geographically, prevent duplicate submissions, and provide total lifecycle transparency for municipal road repairs.

2 Problem Statement

Municipal maintenance departments rely heavily on citizen reports to identify road hazards. However, the current manual reporting pipeline suffers from severe inefficiencies:

- **Data Noise & Invalid Reports:** Citizens frequently submit non-actionable reports, or images that lack actual visual evidence of road damage, wasting human review time.
- **Duplicate Reporting:** Without geographic awareness, multiple citizens often report the exact same pothole, creating redundant database entries and skewing resource allocation.
- **Lack of Lifecycle Transparency:** Once an issue is reported, citizens have no visibility into its status, inspection results, or expected repair time, leading to repeated complaints.

3 Core System Features

To solve these infrastructural challenges, the proposed system will implement the following advanced functionalities:

3.1 1. AI-Powered Automated Verification

Every user-submitted image will be processed in real-time by a deep learning Convolutional Neural Network (CNN). The AI will perform binary classification to determine if the image actually contains a pothole, assigning a confidence score to the prediction. Submissions deemed invalid by the AI are automatically flagged or rejected, drastically reducing administrative bottleneck.

3.2 2. Spatial Data & Duplicate Prevention

The system will capture exact GPS coordinates (Latitude and Longitude) for every approved report. Using spatial database queries, the system will calculate the geographic radius of incoming submissions. If a user attempts to report a pothole within a narrow radius (e.g., 50 meters) of an already existing active report, the system will alert the user and block the duplicate submission.

3.3 3. Centralized Status Tracking & ETA

Every verified pothole enters a strict tracking lifecycle (e.g., *Reported*, *Inspected*, *In Progress*, *Resolved*). The public dashboard will dynamically map these locations. Administrators can update the current status of the repair and assign an Expected Time of Arrival (ETA) for completion, giving citizens total transparency.

4 Technical Methodology

4.1 Artificial Intelligence (Computer Vision)

- **Transfer Learning:** We will utilize a lightweight, pre-trained CNN architecture (such as MobileNetV2) fine-tuned on open-source pothole image datasets to achieve high classification accuracy.
- **Optimization:** The final classification model will be converted to a compressed format (e.g., TensorFlow Lite) for seamless, low-latency integration into the web backend.

4.2 Database Systems & Spatial Engineering

- **Spatial Relational Database:** The system will utilize PostgreSQL. To handle Latitude/Longitude calculations for duplicate prevention, we will implement advanced spatial database queries or rigorous Haversine formula logic.
- **Data Integrity:** The database will be strictly ACID-compliant. It will employ automated internal triggers to update lifecycle timestamps (e.g., `updated_at`) whenever an administrator modifies a pothole's repair status.

4.3 Software Architecture & UI

- **Design Pattern:** The application will be engineered using the Model-View-Controller (MVC) architectural pattern via the Python Flask framework, ensuring a clean separation of database logic, user interface, and AI processing.
- **User Interface:** A responsive, mobile-first frontend will be developed using HTML5, CSS3, and Bootstrap, allowing citizens to easily upload images and GPS data directly from their smartphones.

5 Project Timeline & Milestones

The development lifecycle is structured as an intensive two-week agile sprint to deliver a fully functional Minimum Viable Product (MVP).

Phase	Key Deliverables
Phase 1 (Days 1-3)	Requirements engineering (UML Diagrams), PostgreSQL schema design, and dataset curation.
Phase 2 (Days 4-7)	AI Model training via Transfer Learning, model compression, and core Flask API routing.
Phase 3 (Days 8-11)	Implementing spatial queries, duplicate prevention logic, and integrating the frontend UI.
Phase 4 (Days 12-14)	Cloud deployment, system testing, bug resolution, and final documentation.

Table 1: Two-Week Development Schedule

6 Expected Outcomes

By merging Artificial Intelligence with advanced relational database management and clean software architecture, this project will deliver a highly scalable "smart city" prototype. It will successfully eliminate duplicate administrative work, automate data verification, and provide real-time geographic insights into infrastructure health.